Glass Box: Tasks Left to Complete the Project

LLM hallucination detection (TAI Lab, Sem 5) | Status as of 4 November 2026. Effort figures are rough estimates, not commitments.

1. Where the project stands

Area	State
Working	Windows support (PR #49 merged); terminal demo with token probabilities and a 10-section report; HTML page per answer (press **h**); Qwen 0.5B and 1.5B configs; pinned HaluEval and TruthfulQA data; pair probes for both models (5,000 questions, ECE 0.03 and 0.025).
Main finding	The pair probes score AUROC 0.967 (0.5B) and 0.970 (1.5B) on HaluEval, but answer length alone scores 0.963. With format held equal the 0.5B probe is at 0.50; on TruthfulQA it is at 0.457 with ECE 0.35 and flags none of 121 answers.
Generation probe	Task 7 run (2,400 Qwen answers): held-out AUROC 0.522 on 68 labeled rows (interval 0.39 to 0.67), below answer-text TF-IDF 0.735. Calibration rejected (ECE 0.114 against the 0.10 limit). Not a detector.
Biggest gap	82% of generated answers stay UNVALIDATED (85% on TruthfulQA), so the factuality verdict is missing for most answers and the probe is trained on a biased 18% sample.

2. What the final report must show correctly

These are the terminal and HTML report sections that are still wrong, empty or unsupported, and what each needs.

Report section	Today	Needed before it is correct
[05] Factuality verification	UNVALIDATED for almost every question; the claim catalog has 2 entries	Evidence the claim can be checked against: a larger catalog now, retrieval with provenance later (issue #40)
[07] Probe analysis	Generation probe is experimental and not calibrated; pair probe says 'validated' on its own task only	A probe that beats the text baseline and passes the #18 gate, or an honest negative-result statement in the report
Hallucination probability	NOT AVAILABLE	Shown only if a generation probe passes calibration. Staying unavailable is a valid, honest outcome
[08] Layer-wise analysis	Not available: token spans not captured	Capture token spans and per-layer vectors (ROADMAP Task 8 groundwork)
[09] Early-warning analysis	Not available: no token-level probe, no defined event	Per-prefix evidence labels and the pre-registered event definition (ROADMAP Task 8)
[10] Final conclusion	Limitations list says 'Pairwise probe is not validated for generation factuality' even when a generation probe is loaded (hard-coded text)	Limitations chosen by the loaded probe type, with a test

3. Remaining tasks, in order

Phase A - quick fixes

#	Task	Done when	Rough effort
1	Make the [10] limitations probe-aware (pair, generation, none), so the page and terminal state the right caveat	Unit test per probe type passes; wording matches the loaded probe	1-2 hours
2	Commit and merge outstanding work: frontend PR, pair-probe and artifacts PR, generation and TruthfulQA results branch; add the TruthfulQA row to docs/EXPERIMENTS.md; pin the 1.5B revision in configs/demo-1.5b.yaml	No uncommitted results left; docs match the run records; CI green	2-3 hours
3	HTML page: highlight meaningful digit and name choices instead of punctuation for 'most contested'; show sample-size notes next to scores	Newborn-bones answer highlights the '2' vs '3' step, not the final period	2-3 hours

Phase B - make the results trustworthy

#	Task	Done when	Rough effort
4	Clean-commit reruns (ROADMAP Task 1): pair probe and the generation run	record.json has worktree_dirty false and finished_reproduction true	Overnight compute
5	Length-controlled test for the 1.5B probe; TruthfulQA run with the 1.5B config and probe	Both reports saved and added to EXPERIMENTS.md, whatever the result	1-3 hours compute
6	Labeler audit: hand-label about 100 UNVALIDATED rows with passage evidence; measure how many are truly wrong. If a broader rule is used, write it and check it against the hand labels before opening the test split; then freeze it	Audit table in RESEARCH.md; rule frozen before any new test numbers	1 day

Phase C - make the factuality verdict meaningful (issue #40)

#	Task	Done when	Rough effort
7	Add catalog entries for the demo questions (Austria, Australia, newborn bones, false-premise questions)	Demo answers show FAITHFUL or HALLUCINATED, with the source named	1 hour
8	First #40 PR, no network: evidence record (source, time, hash, quoted span, query), claim IDs and four verdicts, retriever interface with a catalog and a local-snapshot implementation	Fixture tests for missing, stale, contradictory and paraphrased evidence; probe scores never feed the verdict	2-4 days
9	Later #40 work: opt-in online retrieval; small independently reviewed evaluation set	Aggregate numbers published only after the reviewed set exists	1 week or more

Phase D - probe research

#	Task	Done when	Rough effort
10	Resolve the gate threshold mismatch: the #18 gate is frozen at 0.65 but the new pair probe uses 0.95. Record this in #18 as a written amendment before new results	Decision written in the issue; limit values (ECE 0.10, 20 per class) unchanged	1 hour
11	Layer sweep (issue #13) on the generation run; optional 1.5B generation run; larger reviewed label set from step 6	Reports saved for each variant against the same frozen split	1-2 days
12	Success test: probe beats TF-IDF with the interval clear of it, test ECE at most 0.10, at least 20 per class in validation and test, gate #18 passes	Either a validated probe, or a clearly written negative result in the final report	Decision point
13	Only if step 12 passes: Brier score and interval estimates for calibration	Reliability report with intervals; null when support is too small	1-2 days

Phase E - report sections [08] and [09] (stretch)

#	Task	Done when	Rough effort
14	Capture token spans; add per-prefix evidence labels; evaluate early warning (ROADMAP Task 8). Activation intervention (Task 9) is out of scope	Lead time stays null unless per-prefix events are stored	1 week or more

Phase F - polish and delivery

#	Task	Done when	Rough effort
15	Remove the Ruff E501 ignore (Task 3); add a type-checker gate (Task 10); choose a license (Task 6, maintainer); add the CRLF dataset note to docs/DATASETS.md	ruff, tests and CI pass; README matches the repo	1 day
16	Final write-up and demo: findings table (length shortcut, controlled test, TruthfulQA, generation probe), limitations, demo script of 5-6 questions with screenshots of the HTML page	Report and demo run end to end on a clean checkout	2-3 days

4. Rules to keep

- Do not change the ECE limit (0.10), the 20-per-class minimum or any frozen threshold after seeing results.
- Never treat UNVALIDATED answers as hallucinations; missing evidence is not a hallucination.
- Keep pair-protocol and generation-protocol metrics separate; always show the answer-length and TF-IDF baselines next to a probe's AUROC.
- Do not commit large prediction files (predictions.json 8 MB, review_queue.json 5 MB); share them another way.
- Hallucination probability appears only when a generation probe passes calibration. Do not add a path around that gate.

5. If time is short

Minimum honest finish: tasks 1, 2, 3, 7 and 16. The project then ships a working demo, a correct report, and a clearly argued negative result on hidden-state probing. **Best use of extra time:** task 6 (it shows how much of the 82% is truly wrong), then task 8 (a first real factuality check). Tasks 4, 5 and 10 to 13 add rigor; tasks 14 is a stretch.